
INTEROFFICE COMMUNICATION

TO: Brandi Wheeler, Project Manager, Cadillac District Office
Remediation and Redevelopment Division

FROM: Ian Halbeisen, Geologist, Geological Services Section
Remediation and Redevelopment Division

DATE: September 4, 2025

SUBJECT: Cadillac Area Per- and Polyfluoroalkyl Substances (PFAS) Investigation
Wexford County, Location ID: MISC0426, GSS Job #2413
Monitor Well Installation

This memorandum summarizes the work requested by the Department of Environment, Great Lakes, and Energy (EGLE), Remediation and Redevelopment Division's (RRD's), Cadillac District Office for the subject site. The Cadillac District Office requested RRD's Geological Services Section (GSS) install monitor wells for the investigation of PFAS and to establish groundwater flow direction. GSS conducted the fieldwork on July 14-16, 2025.

This memorandum includes the following:

- Site Map (Figure 1)
- Groundwater Flow Direction Map (Figure 2)
- Global Positioning System (GPS) Data (Table 1)
- Groundwater Elevation Data (Table 2)
- Monitor Well Logs (Appendix A)

SITE BACKGROUND

The site is located in Sections 27 and 28, T22N-R09W, Haring Township, Wexford County, Michigan (Fig 1). The monitor well locations were previously selected by the district project manager.

MONITOR WELL INSTALLATION

On July 14, 2025, GSS arrived at the site and installed three monitor wells (GSS-1, GSS-2, and GSS-3) (Appendix A). Staff installed GSS-1 south of the parking lot for the soccer fields. The parking lot is northeast of the intersection of Plett Road and 13th Street/36 Road. GSS-2 was installed at the Wexford County Fairgrounds. The well is inside the fairground fence just east of the south entrance gate. GSS-3 was installed west of the intersection of Bell Street and Plett Road (43 Road), across the road from the residence at 801 Bell Street.

The geology is mostly fine to coarse sand, light to medium brown. While drilling GSS-2, staff encountered gravel at 45 feet. At GSS-3, gravel was encountered at 25 feet.

The GSS surveyed the monitor wells with GSS-3 having an assumed top of casing elevation of 100.00 feet. Staff recorded the monitor well locations using GPS (Table 1).

On July 17, 2025, GSS measured the static water levels (Table 2). Groundwater flow direction is northwesterly (Fig 2). Per district's request, GSS did not collect any soil or water samples.

If you have any questions, contact me at 517-582-3071.

A handwritten signature in cursive script, appearing to read "Jim Halboisen".

cc/att: Aaron Berndt, EGLE
Brett Coulter, EGLE
Scott Denstadt, EGLE



LEGEND

● Monitor Well



DATUM - NAD83
PROJECTION: MICHIGAN GEOREF
NORTHING AND EASTING COORDINATES (IN METERS)
ARE IN CORNERS OF MAP

AERIAL PHOTO SOURCE: MISAIL Imagery

0 20 40 80 Meters

0 75 150 300 Feet

1:3,600



PFAS INVESTIGATION LOCATION ID MISC0426

CITY OF CADILLAC, WEXFORD COUNTY
T22N R9W SECTION 27 & 28

Site Map

GEOLOGIST
Ian Halbeisen
Geological Services
Section

Remediation and
Redevelopment
Division

EGLE
Michigan Department of
Environmental Quality

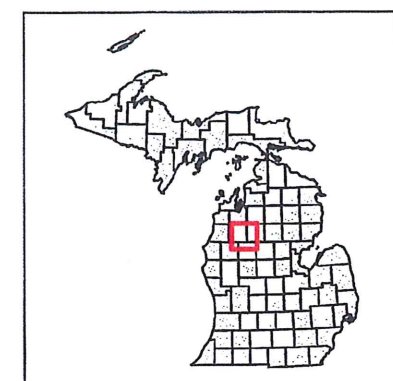
CREATION DATE
September 2025

FIGURE 1



LEGEND

● Monitor Well



DATUM - NAD83
PROJECTION: MICHIGAN GEOREF
NORTHING AND EASTING COORDINATES (IN METERS)
ARE IN CORNERS OF MAP

AERIAL PHOTO SOURCE: MISAIL Imagery

0 20 40 80 Meters



0 75 150 300 Feet



1:3,600



PFAS INVESTIGATION LOCATION ID MISC0426

CITY OF CADILLAC, WEXFORD COUNTY
T22N R9W SECTION 27 & 28

Groundwater Flow Direction Map

GEOLOGIST
Ian Halbeisen
Geological Services
Section

Remediation and
Redevelopment
Division



CREATION DATE
September 2025

FIGURE 2

Title	Latitude	Longitude	Northing	Easting	Max_PDOP	Corr_Type	GPS_Date	Unfilt_Pos	Horz_Prec	Std_Dev
GSS-1	44.267637099	-85.396250606	413083.233	548028.999	1.5	L1L2 Postprocessed Carrier Float	7/15/2025	91	0.1	0.019828
GSS-2	44.267175793	-85.404237039	413027.294	547391.972	2.3	L1L2 Postprocessed Carrier Float	7/15/2025	98	0.1	0.022327
GSS-3	44.274006573	-85.398235533	413789.516	547865.346	6.2	L1L2 Postprocessed Carrier Float	7/15/2025	110	0.2	0.434463

Monitor Well	Well Depth (ft)	Top of Casing Elevation	Ground Elevation	Static Water Level 07/17/25	Groundwater Elevation 07/17/25
GSS-1	51	91.56	91.9	15.37	76.19
GSS-2	50	103.10	103.3	29.25	73.85
GSS-3	50	100.00	100.4	25.9	74.10

APPENDIX A

Cadillac Area PFAS Investigation, Wexford County
Location ID: MISC0426

Monitor Well Logs

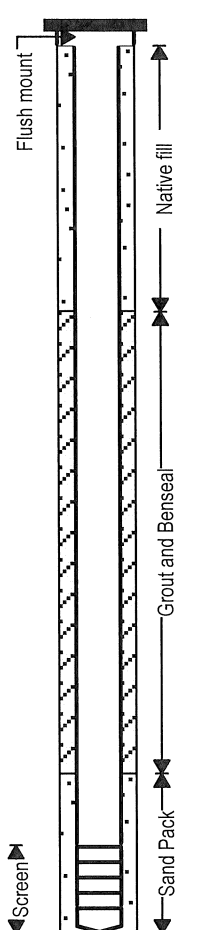
SITE: PFAS INVESTIGATION

BOREHOLE LOG

COUNTY: WEXFORD
TOWNSHIP: HARING
TOWN: T22N
RANGE: R9W
SECTION: 27
LOCATION DESCRIPTION: South of parking lot for soccer fields, north of Clam River.

DATE: July 15, 2025
DRILLER: Scott Densteadt
GEOLOGIST: Ian Halbeisen
DRILL METHOD: Hollow stem augers
TOTAL DEPTH: 51 feet

LOCATION ID: MISC0426

WELL CONSTRUCTION	LITHOLOGIC LOG	DESCRIPTION	DEPTH	SAMPLE ID	SAMPLE TYPE	FIELD SCREENING RESULTS	PID ppm
							01000
		Grd. SAND Fine grained sand, some loam with occasional pebbles, dry to moist, dark to medium brown	0				
		LOAMY SAND Fine grained sand, loamy, dry to moist, dark brown to black	5				
		SAND Medium grained sand, moist, medium brown	10				
		SAND Fine grained sand, moist to wet, light brown	15				
		SAND Coarse grained sand, wet, light brown to light grayish brown	20				
			25				
			30				
			35				
			40				
			45				
			50				
		E.O.B.	55				

VERTICAL DATUM: Surveyed from GSS-3 (TOC), assumed value of 100.00 feet.
GRD. ELEVATION: 91.9 feet
T.O.C.: 91.56 feet
S.W.L.: 15.37 feet
CASING: 2-inch PVC
SCREEN: 2-inch PVC 5 foot long
WELL DEPTH: 51 feet (TOC)
COMPLETION NOTES: Sand pack to 42 feet, grout with 6 sx Enviroplug and 7 sx of Holeplug to 16 feet, filled rest of hole with sand from boring.

LATITUDE: 44.267637099
LONGITUDE: -85.396250606
DATUM: Michigan GeoRef
NORTHING: 413083.233
EASTING: 548028.999

SITE: PFAS INVESTIGATION

BOREHOLE LOG

COUNTY: WEXFORD

TOWNSHIP: HARING

TOWN: T22N

RANGE: R9W

SECTION: 28

LOCATION DESCRIPTION: Inside gate at fairgrounds on right hand side.

DATE: July 15, 2025

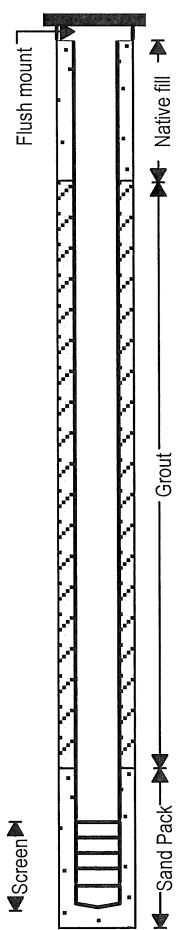
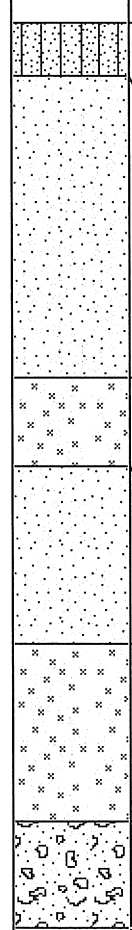
DRILLER: Scott Densteadt

GEOLOGIST: Ian Halbeisen

DRILL METHOD: Hollow stem augers

TOTAL DEPTH: 51 feet

LOCATION ID: MISC0426

WELL CONSTRUCTION	LITHOLOGIC LOG	DESCRIPTION	DEPTH	SAMPLE ID	SAMPLE TYPE	FIELD SCREENING RESULTS	PID ppm 01000
		Grd. LOAMY SAND Fine grained sand, loamy, dry to moist, dark brown to black SAND Fine grained sand, few fines, dry to moist, light brown SAND Fine to coarse grained sand with pebbles, moist, light grayish brown SAND Fine grained sand with occasional pebbles, moist to wet, light to medium brown SAND Fine grained sand increasing pebbles and occasional clay lenses, wet, light grayish brown Fine to medium grained sand, gravelly with coarser gravel, wet, medium brown to grayish brown E.O.B.	0 5 10 15 20 25 30 35 40 45 50 55				

VERTICAL DATUM: Surveyed from GSS-3 (TOC), assumed value of 100.00 feet.

GRD. ELEVATION: 103.3 feet

T.O.C.: 103.10 feet

S.W.L.: 29.25 feet

CASING: 2-inch PVC

SCREEN: 2-inch PVC 5 foot long

WELL DEPTH: 50 feet (TOC)

COMPLETION NOTES: Sand pack to 42 feet, grout with 6 sx Enviroplug to 9 feet, filled rest of hole with sand from boring.

LATITUDE: 44.267175793

LONGITUDE: -85.404237039

DATUM: Michigan GeoRef

NORTHING: 413027.294

EASTING: 547391.972

SITE: PFAS INVESTIGATION

BOREHOLE LOG

COUNTY: WEXFORD

TOWNSHIP: HARING

TOWN: T22N

RANGE: R9W

SECTION: 28

LOCATION DESCRIPTION: South of Bell Avenue, across the street from 801 Bell Avenue.

DATE: July 16, 2025

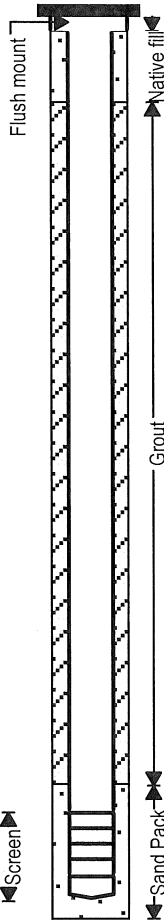
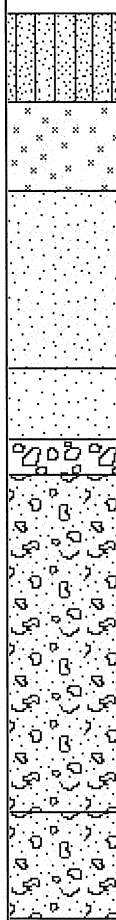
DRILLER: Scott Denstead

GEOLOGIST: Ian Halbeisen

DRILL METHOD: Hollow stem augers

TOTAL DEPTH: 51 feet

LOCATION ID: MISC0426

WELL CONSTRUCTION	LITHOLOGIC LOG	DESCRIPTION	DEPTH	SAMPLE ID	SAMPLE TYPE	FIELD SCREENING RESULTS	PID ppm 0 1000
		Grd. LOAMY SAND Fine grained sand, loamy with some gravel, dry to moist, dark reddish brown to black SAND Fine to medium grained sand, dry to moist, light to medium brown SAND Fine grained sand, moist, light brown SAND Fine grained sand with some gravel, moist, light brown GRAVEL ZONE Mostly gravel based on drillers report. SAND Fine to medium grained sand, gravelly, moist to wet, light to medium brown SAND Fine to coarse grained sand with fine gravel, wet, light grayish brown E.O.B.	0 5 10 15 20 25 30 35 40 45 50 55				

VERTICAL DATUM: Surveyed from GSS-3 (TOC), assumed value of 100.00 feet.

GRD. ELEVATION: 100.4 feet

T.O.C.: 100.00 feet

S.W.L.: 25.90 feet

CASING: 2-inch PVC

SCREEN: 2-inch PVC 5 foot long

WELL DEPTH: 50 feet (TOC)

COMPLETION NOTES: Sand pack to 43.5 feet, grout with 6 sx Enviroplug to 5 feet, filled rest of hole with sand from boring.

LATITUDE: 44.274006573

LONGITUDE: -85.398235533

DATUM: Michigan GeoRef

NORTHING: 413789.516

EASTING: 547865.346